

Litterature review

Social media use has increased markedly the last 2 decades [1]. As social media has become a permanent part of life, protecting children from the potential harms that can affect them when using social medias is increasingly important. Some of the benefits from the use of social media are promoting diversity, fostering friendships, keeping contact over greater distances and improving communication skills [2].

There is an association between the use of social media and self-esteem and body image. Cyberbullying is the most common online risk for all young people, and the effects of it can include depression and anxiety among others [3]. Several studies highlight risks of social media use among children. Some of them involve personal safety, privacy concerns and access to inappropriate material. It was also proposed that greater digital competence could increase exposure to online risks [4]. Although mobile phones can help improve adolescents' health and psychosocial well-being, a study found that late-night mobile phone use can be associated with sleep problems [1].

The research the area of children and social media is insufficient because of a lack of guidance for parents and teachers[3]. The best thing to prevent children from the risks of using social media, is to raise public and medical awareness. If health professionals, caregivers and developers collaborate, they can guide each other to find the best solutions. Responsible media use is important for children to learn to safeguard their privacy, while being guided by responsible parents [4]. By using appropriate website design the risks of social media use can be reduced, especially by adding safety features. This will not be as easy to do in social media sites used by adult, but in software aimed at kids it should be implemented[3]. These findings provide a foundation for evaluating existing social media platforms aimed at children, to try and identify how safety and social features both be increased at the same time.

Software review

In this software review the social media platforms ZigaZoo, Roblox, Woozworld and Zoomerang will be examined. Information about them was gathered primarily from the “about us” sections on their official website.

Many of the platforms focus on the creative, particularly video-based activities. Both ZigaZoo and Zoomerang has some form of video concept where kids can edit and share videos answering to prompts or challenges [5], [6]. Roblox and Woozworld has more of an “online world” concept with avatars and chat based activities [7], [8].

All the platforms included in this review implement some moderation. Roblox allows users under 13 to access the platform, but those users have stricter controls and filters and cannot send private messages to other users [8]. ZigaZoo requires parental consent to setup an account, and has human moderators to ensure safety [5]. Zoomerang has a 13-year age limit [5]. Woozworld is intended for children age 16 or older, children below 16 years need parental consent and if they are below 13 they get some restrictions [8].

Zoomerang allows in-game purchases and has ads and possibilities for subscription. If you are above 13 years, it is possible to make money if you get enough followers. The software focuses on videos and easy video editing including music, filters and trends. There are no open feed, but there is a follower-system and you cannot share the videos outside the app [6].

Roblox is a software where the users make the game content with Roblox-studio. The users can have friends and followers and can chat with both private messages and in game. Robux is an in-game currency, and you can purchase items in the game with it [8].

Woozworld is an avatar based social world, where the users connect with their friends and followers through events and games. It is possible to design clothing, rooms and environments, and join competitions. The users can get an in-game currency and purchase cosmetic items. The software has an report function for inappropriate content, and has chat filtering in controlled chat-rooms [7].

ZigaZoo is a video-sharing social platform built around creation. The users create videos responding to prompts given by the platform. Users can not comment or message each other, but they can follow each other and react to videos with their own video or a reaction [5].

Brainhub

The target user group for 'Brainhub' is children aged 10-14. Most social networks have an age limit of 13, but since this platform does not include direct messaging and focuses on solving daily puzzles and sharing achievements on a public feed, the minimum age was lowered to 10 years. The main goal for this platform is to make time children spend online more meaningful by encouraging cognitive activity, rather than scrolling.

The main use cases are children using the platform to complete daily challenges and being able to share the fact they completed puzzle to their friends. The goal for this social platform is to make puzzles more fun and social, and easier to complete, instead of scrolling and not using the brain when being online.

Use-case 1: Completing daily sudoku challenge

User navigates to sudoku page with button – reads the rules for sudoku – solves sudoku – gets confetti and a popup – chooses to share activity to feed.

Use-case 2: adding a book to the library

User navigates to library with button - clicks add book– fills in input fields – clicks add– gets popup if the user wants to share the activity – user shares activity

Use-case 3: answer daily quiz

User navigates to quiz page with button – reads the question – thinks of an answer – presses the 'Show answer' button to see if they were right – after finishing the quiz the user can press the score button that matches their score, and can share it with their friends

The main design features that are implemented are an activity feed to show feed-updates from friends, including their avatar and reactions to click on, a navigation grid on the home page to access each module, popups for confirmation and sharing actions, and consistent icon-based navigation, including a "back" button on the top left and a profile button on the top right. The user can solve a daily sudoku, a daily quiz and add/remove books in their own online library. These features follow established user interface(UI) conventions to make it more accessible and user friendly.

When playing sudoku, the numbers are placed under the sudoku-board and are using the method, select a tile and place a number, which are used by several other online sudokus. This also makes it easier for the user to not make mistakes, because it is a two-step process. The user can also toggle the rules to read whilst playing the sudoku. To reduce the gulf of

evaluation when playing sudoku, a wrong move makes the tile blink red, and the mistakes counter add +1 to the count. Both the selected tile and number have a darker colour than the others to show that they have been pressed. When the user completes the sudoku, there is confetti and a popup which tells the user good job and gives them the choice to close or share the news.

The library page also follows established UI patterns so reduce the gulf of execution. The different books are showed with individual cards with information about title, author and a cover the user can chose themselves. The button for adding a new book is big and in the centre of the page. When you add a book, you get input fields to fill in with information about the book. This is very normal for this type of use, but to help the user reduce the gulf of evaluation the input fields have prefilled text to show what to write where. The user also gets an opportunity to share the news to their friends, with the same function as on the sudoku page. Each book-card has a read/marked as read-button. This button does not change colour, but it has text that tells the user what the button does very clearly. Above to the right of the book-cards there is a book-counter that goes up when you add a book, also showing the user feedback of their actions. All the pages have a go back-button on the left top of the page, and a profile button on the top-right. This follows conventions and established UI design patterns.

When using the home page, the design principle about a conceptual model is used to create an understanding and a feeling of control. By making the interface reflect an appropriate model, the user can learn it with less guidance and fewer errors [9]. This enhances the tolerance for error, and the discoverability, and it is very easy to move to the different modules/pages because the focus on the screen are buttons to the different pages the user can navigate to. This also shows good mapping[10], [11].

The buttons are all easy to read and are quite big, especially on the home page, and that makes them good signifiers. They have either a hover-effect or a moving effect when hovering so that it is easy to get feedback when clicking, this is a signifier for the affordance of clicking the button to make an action. To make the buttons even more accessible a feedback sound or vibration when clicking would make the design communicate the necessary information effectively[10].

There are also popup-notifications when the user press share, giving feedback which say “this has been shared to your feed”. There is also a cancel button to let the user decide if they want to regret clicking the share button, making errors easier to handle. If a user wants to add a

book but forgets to fill in one of the input fields, they get a message telling them about the mistake and asking them to try again. Both error-prevention and clear feedback is important to create a good user experience [11], [12].

The used fonts are easy to understand, which is important when it is a platform for children. They often have different reading abilities, and an easy font is inclusive. The different reactions of the feed entries have icons instead of text, so it is easier to understand the reaction the user is sending, and the profile button is the users avatar which may not be very intuitive at first glance, but the profile button placement is very standard. The layouts for the different modules are very similar, keeping high consistency and not increasing the users cognitive load[12]. This will hopefully make the use of the design more efficient and comfortable with a minimum of fatigue[10].

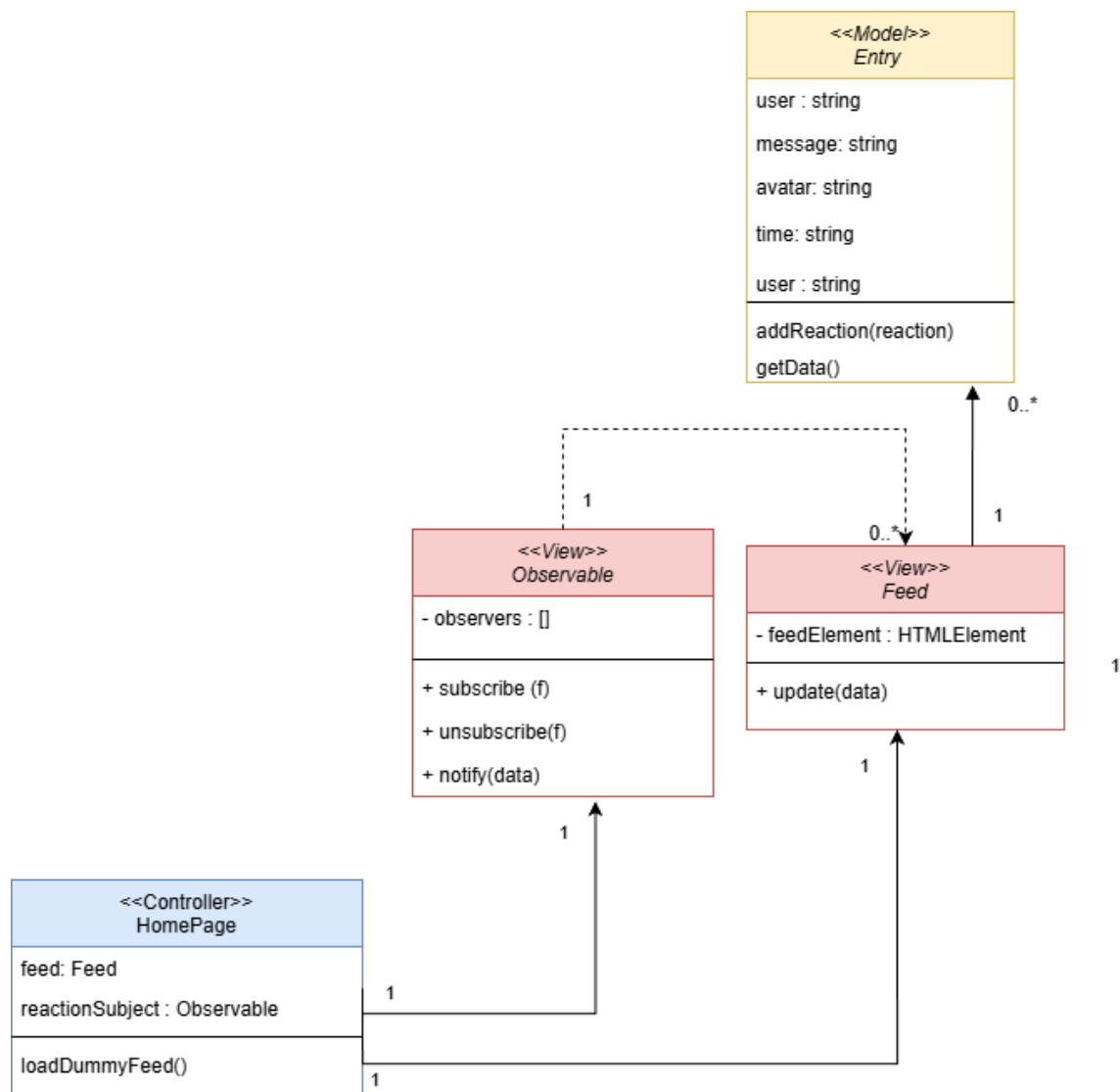
The software architecture is model view controller (MVC). The MVC is appropriate for the social network because it keeps the interface simple and consistent, and it allows mini-apps like sudoku and the library to share a structure, and it is easy to expand with more modules like a memory game. The MVC architecture is typically deployed at the level of a single application. In this architecture, the model stores the data presented by the user interface, the view visualizes the data in the model, and the controller makes decisions about how to handle user input and update the model. The UI is highly interactive and frequently updated with feed notifications, popups and counters. The components like the Book class and feed system are modular and reusable and therefore supporting future expansions. It is also easier to test each part individually because of the separation of the different parts and responsibilities. MVC [13].

The software implements the Observer design pattern to provide reactive UI feedback, such as popups counter, which is well suited for children that may need a bit more reinforcement than adults. A major feature of the platform is the activity feed, where users can share achievements and receive reactions on the posts, making the Observer pattern a good fit. Initially, the plan was for the library, quiz and sudoku module to notify the feed whenever a new activity was shared. However, this implementation proved difficult to implement fully. Therefore, a simplified Observer pattern was implemented on the home page to handle post reactions, while the architecture allows for future expansion to other models. The system also uses the Factory pattern when creating a new book, ensuring that all book objects share the same structure. This approach makes room for future extensions, such as implementing

Strategy patterns for filtering or sorting books. Currently books can be sorted alphabetically, but the architecture allows for more complex filtering to be added later[14].

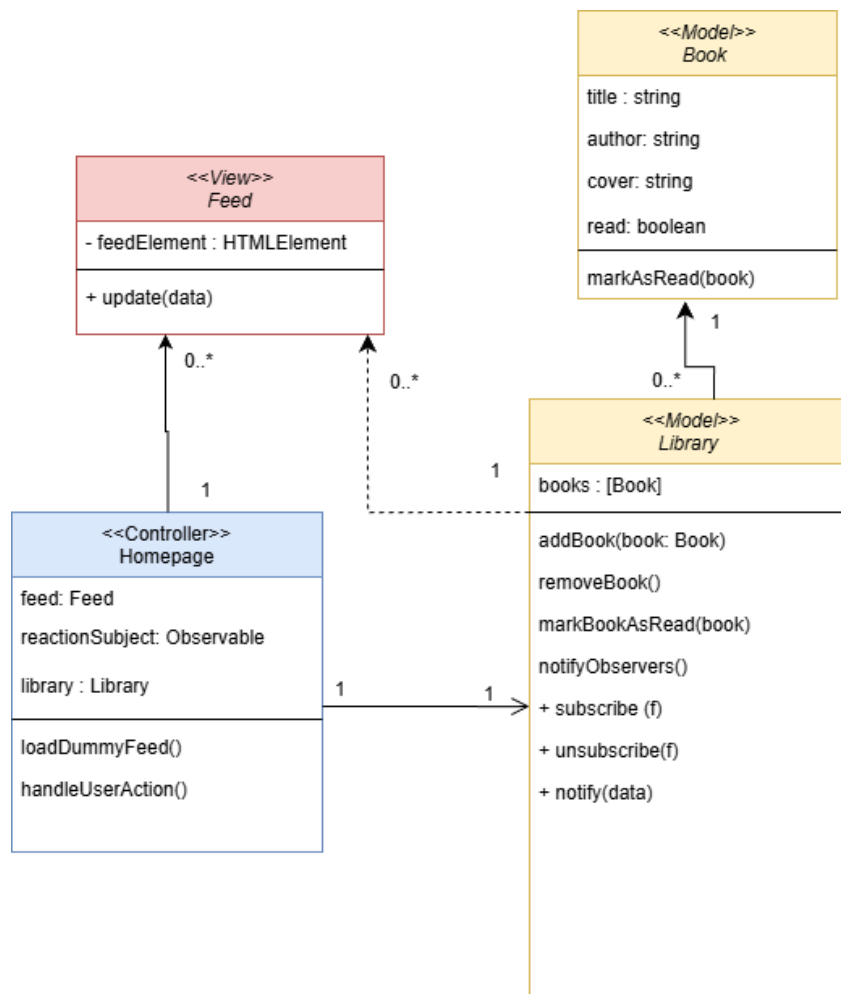
UML diagrams:

Figure 1: Dataflow in main.js / children_social.html



Description: **HomePage** is the controller and initialises feed and observer subscription. **Feed** subscribes to observable and makes entry-elements which represents user activities, and sends reactions to **Observable** (`reactionSubject`). **Observable** notifies all the subscribers when a reaction happens, and tells the UI. **Entry** represent the posts in the feed, and has data about the user, avatar, message and the different reactions.

Figure 2: Dataflow in library.js / library.html to



Description: Here the library works as the systems model component and saves all the Book-objects and is the users digital library/bookshelf. Book contains info about title, author, cover and if it is read. Library offers methods for adding books, removing books, update the read-status and fetch saved data. The controller makes decisions about how to handle the user input and update the model. In a full implementation library would act as an observable, in an observer pattern, where changes in the model would create a notify call to all subscribers, here Feed, and they would update their interface when changes occur. Even though the code does not do exactly this, the diagram shows how the system is structured to use the pattern for further use.

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